

Chapter 1.2

Introduction to Dart

Introduction

Dart is an open-source, general-purpose programming language developed by Google

Applications:

- Web development
- Mobile development
- Server-side development

Introduction

Key features:

1. Object-oriented
2. Strongly typed
3. Fast and efficient
4. Modern features

Structure

```
// This is a comment
```

```
void main() {
```

```
    // This is the main function, program starts here
```

```
    print("Hello, world!");
```

```
}
```

Variable

Data Types

Numbers:

`int`: Stores whole numbers (integers).

`double`: Stores numbers with decimals (floating-point numbers).

`num`: Supertype of both `int` and `double`.

Variable

Declaration Keywords

// Declares an integer variable, needs initialization before use

```
int age;
```

// Declares a double variable

```
double pi = 3.14159;
```

// Declare multiple variables of the same type in a single line

```
int x = 5, y = 10;
```

Variable

Declaration Keywords

```
// Nullable double, initially null
```

```
double? area = null;
```

Variable

Data Types

Text:

String: Stores sequences of characters (text).

Variable

Declaration Keywords

```
// String type with initial value
```

```
String name = "Alice";
```

```
// var infers String based on assignment
```

```
var greeting = "Hello";
```

```
// var infers String based on assignment
```

```
var flybyObjects = ['Jupiter', 'Saturn', 'Uranus', 'Neptune'];
```

Variable

Data Types

Logical:

`bool`: Stores true or false values (Boolean).

Variable

Declaration Keywords

```
bool isLoggedIn = true;
```

```
bool isNightTime = false;
```

Basic Structures

```
if (year >= 2001) {  
    print('21st century');  
} else if (year >= 1901) {  
    print('20th century');  
}
```

Basic Structures

```
for (int month = 1; month <= 12; month++) {  
    print(month);  
}
```

```
for (final object in flybyObjects) {  
    print(object);  
}
```

Basic Structures

```
while (year < 2016) {  
    year += 1;  
}
```

Functions

```
bool isNoble(int atomicNumber) {  
    return _nobleGases[atomicNumber] != null;  
}
```

Functions

```
int fibonacci(int n) {  
    if (n == 0 || n == 1) return n;  
    return fibonacci(n - 1) + fibonacci(n - 2);  
}
```

```
var result = fibonacci(20);
```


Functions

```
bool isNoble(int atomicNumber) => _nobleGases[atomicNumber] !=  
null;  
  
flybyObjects.where((name) => name.contains('turn')).foreach(print);
```

Find Out More

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